

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 18

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Write a program that creates an array of 1,000 random integers and finds the largest value using a parallel loop.
2. Write a program that creates three arrays A, B, C of size 10,000 and computes vector addition: $C[i] = A[i] + B[i]$. Use a parallel loop a temporary variable inside the loop holds the sum of $A[i]$ and $B[i]$ before assigning it to $C[i]$.
3. Write a program that estimates the value of pi. Run 1,000,000 iterations. In each iteration, generate two random numbers, check the condition, and increment a hit counter if it's inside the unit circle. Use `firstprivate` for your random number seed to ensure each thread starts its sequence differently, and reduction for the hit counter.

- **Welcome: Sean**

- [LogOut](#)

